



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

COMBINED SCIENCE

PAPER 3 Practical Test

4003/3

1 hour 30 minutes

SPECIMEN PAPER

ADVANCE INSTRUCTIONS TO SUPERVISORS

Great care should be taken that the confidential information in these instructions does not reach the candidates either directly or indirectly.

Instructions for the Supervisor.

These instructions should be treated and handled as examination live material.

Prior to the examination, the Supervisor should ensure that all apparatus provided for the candidate as listed in these instructions, are in good working condition.

The Supervisor is **NOT** allowed to access the Question paper before the examination **Date** and **Time**.

These Advance Instructions consist of 4 printed pages.

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Instructions for preparing apparatus and supervising the Practical Test

- 1 In addition to the usual equipment of a Science laboratory, candidates will require the apparatus specified in these instructions. Each candidate can spend 45 minutes using the apparatus for each of Questions 1 and 2.

Enough apparatus must be provided to ensure that

- (a) no candidate is kept waiting at the end of the allocated time for each question,
- (b) spare apparatus is immediately available in the event of failure or breakage of any item.

- 2 If a candidate is unable to proceed with any of the experiments **due to failure to use the apparatus**, the Supervisor is expected to give sufficient help to **enable the candidate to use the apparatus**. In giving help, the following regulations should be followed:

- (a) No help may be announced to the candidates as a whole
- (b) Candidates should be told that the Examiners will be informed of any help given
- (c) **A report must be made of any assistance given to any candidate, identified by name and candidate number**

Supervisors may find it useful to make the following announcement to the candidates at the start of the examination. **‘The examiners do not want you to waste time when you are unable to use the apparatus. Any candidate who is unable to get on with an experiment due to failure to use the apparatus may ask for help. I will report to the Examiners any help given and some marks may be deducted.’**

NB: Help given by the Supervisor should be limited to the use of apparatus and should be communicated through the Supervisor’s report.

- 3 Page 8 should be used to provide information about any help given to candidates. Pages 7 and 8 should be detached and attached to the ‘Supervisor’s Results’. **Details should be given if the apparatus used differ(s) in any significant way from that specified in these Instructions.**
- 4 In addition to the usual equipment of a science laboratory, each candidate will require the apparatus specified overleaf. If a candidate breaks any of the apparatus, or loses any of the material supplied, the matter should be rectified and a note be made in the report.
- 5 **On the day of the examination, the supervisor is asked to perform the experiments in questions 1 and 2 and to record the results on a spare copy of the question paper clearly labelled “Supervisor’s Results”, followed by the Centre Number. This must be enclosed with the scripts. Unless this is done, candidates may be unavoidably penalised.**
- 6 In large centres where scripts are despatched in more than one envelope, a copy of the “Supervisor’s Results” should be sent with the scripts for each group.
- 7 **The Supervisor should not communicate directly to candidates but should do so through the invigilator unless it is a matter of safety.**

General Information for the Supervisor

The Supervisor is the Science teacher who will be in charge of the examination.

The Supervisor may improvise on the apparatus to enable the candidates to carry out the experiments for example, 10 ml measuring cylinders may be used instead of graduated syringes.

Also, instead of providing candidates with boiling tubes, the supervisor may provide test tubes if the heating is done through a water bath.

The experiments done by candidates during the practical examination can be carried out in a well-ventilated ordinary classroom. The Supervisor, for example, may provide communal spirit or gas burners, may use buckets half filled with water as basins to pour away waste materials after rinsing, may provide water in plastic bottles so that candidates may use it for rinsing etc.

Where the practical is not qualitative in nature, water may be used for the dilutions instead of distilled water.

The Supervisor should communicate any improvisations made that might negatively affect the candidates through the Supervisor's Report.

NB: Candidates may be allowed justifiable extra time to a maximum of 10 minutes if they are disadvantaged by sharing apparatus.

The Practical Examination may be administered in sessions if there is limited equipment.

When a centre has more than one session, all the candidates should be at the centre before the examination starts, that is, before 9.00 AM.

Only the candidates yet to sit for the examination should be quarantined with teacher supervision.

The quarantined candidates should NOT have access to cell phones.

Each candidate requires**Question 1**

100 cm³ of dilute sulphuric acid (1.0 mol/dm³)

100 cm³ 1.0 M sodium hydroxide solution

phenolphthalein indicator

a conical flask

a 25 cm³ pipette

a pipette filler

a burette

a filter funnel

a dropper

a white tile

retort stand and clamp

NB: The Supervisor should mount the empty burette.

Question 2

access to an electric circuit.

Each circuit requires the following:

connecting wires

ammeter

voltmeter

3 × 1.5 V cells

switch

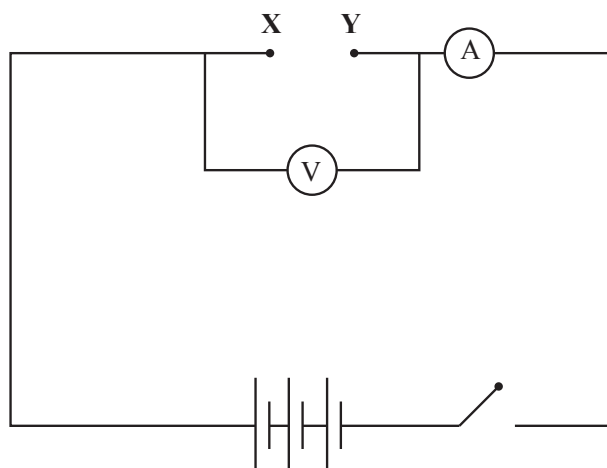
10 cm constantan wire SWG 24 labelled E

10 cm constantan wire SWG 26 labelled F

10 cm constantan wire SWG 28 labelled G

10 cm constantan wire SWG 32 labelled H

NB: The supervisor should set the circuits as shown in the diagram.



Place crocodile clips on points **X** and **Y**.